

SalesGPT Framework

An Asynchronous Dual-Track Agentic System for Autonomous Lead Qualification

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1. Abstract

SalesGPT is an innovative, asynchronous artificial intelligence system designed to address the critical challenge of lead leakage in enterprise sales environments.

By implementing a dual-track architecture that decouples response latency from intelligence depth, the system simultaneously delivers real-time customer interactions via Retrieval-Augmented Generation (RAG) while conducting background behavioral analysis using the BANT framework.

2. INTRODUCTION

The Enterprise Sales Gap

Modern B2B sales organizations face a critical inefficiency: the inability to rapidly and accurately qualify high-value prospects from high-volume conversational data.

Traditional systems either provide instant, generic responses or require slow, manual analysis—resulting in a loss of 30-40% of qualified leads.

The Proactive Engine

SalesGPT transforms a simple customer support channel into a 24/7 proactive sales intelligence engine.

It acts as a fully autonomous layer that engages prospects instantly, analyzes their purchasing intent behind the scenes, and alerts human sales representatives only when a lead is highly qualified.

3. PROBLEM STATEMENT

-  **Temporal Mismatch:** Response speed requirements directly conflict with the computational depth required for complex behavioral analysis.
-  **Lead Leakage:** High-quality, enterprise prospects often go unrecognized amid voluminous casual inquiries and generic support requests.
-  **Manual Overhead:** Sales teams must manually review raw conversation transcripts to identify qualified leads, wasting valuable selling time.
-  **Context Loss:** Extracted buyer signals (contact information, technical needs, budget indicators) remain scattered across disparate chat systems rather than unified in the CRM.

4. EXISTING SYSTEM

Rule-Based Chatbots & Basic RAG

Current enterprise chatbots rely heavily on rigid decision trees or synchronous single-layer Large Language Models.

These systems are fundamentally passive; they respond accurately to user queries but fail to proactively analyze the psychological or business intent behind those queries.

The Speed vs. Intelligence Trade-off

Forcing a single LLM to simultaneously answer a technical question and analyze BANT (Budget, Authority, Need, Timeline) creates severe latency.

This results in degraded user experiences (multi-second delays) or forces companies to abandon automated lead scoring entirely.

5. PROPOSED SYSTEM



Fast Track (Synchronous)

Handles immediate user interaction via RAG. Queries pgvector databases and utilizes Llama-3.3-70B to generate highly accurate, context-aware responses in under 1.5 seconds.



Slow Track (Asynchronous)

Executes entirely in the background. A 120B parameter Judge Agent analyzes the conversation transcript for BANT signals without blocking the user's chat experience.



Automated CRM Enrichment

Parallel extractor agents pull contact data, update the real-time Kanban pipeline, and auto-draft personalized email follow-ups based on detected purchasing intent.

6. LITERATURE REVIEW

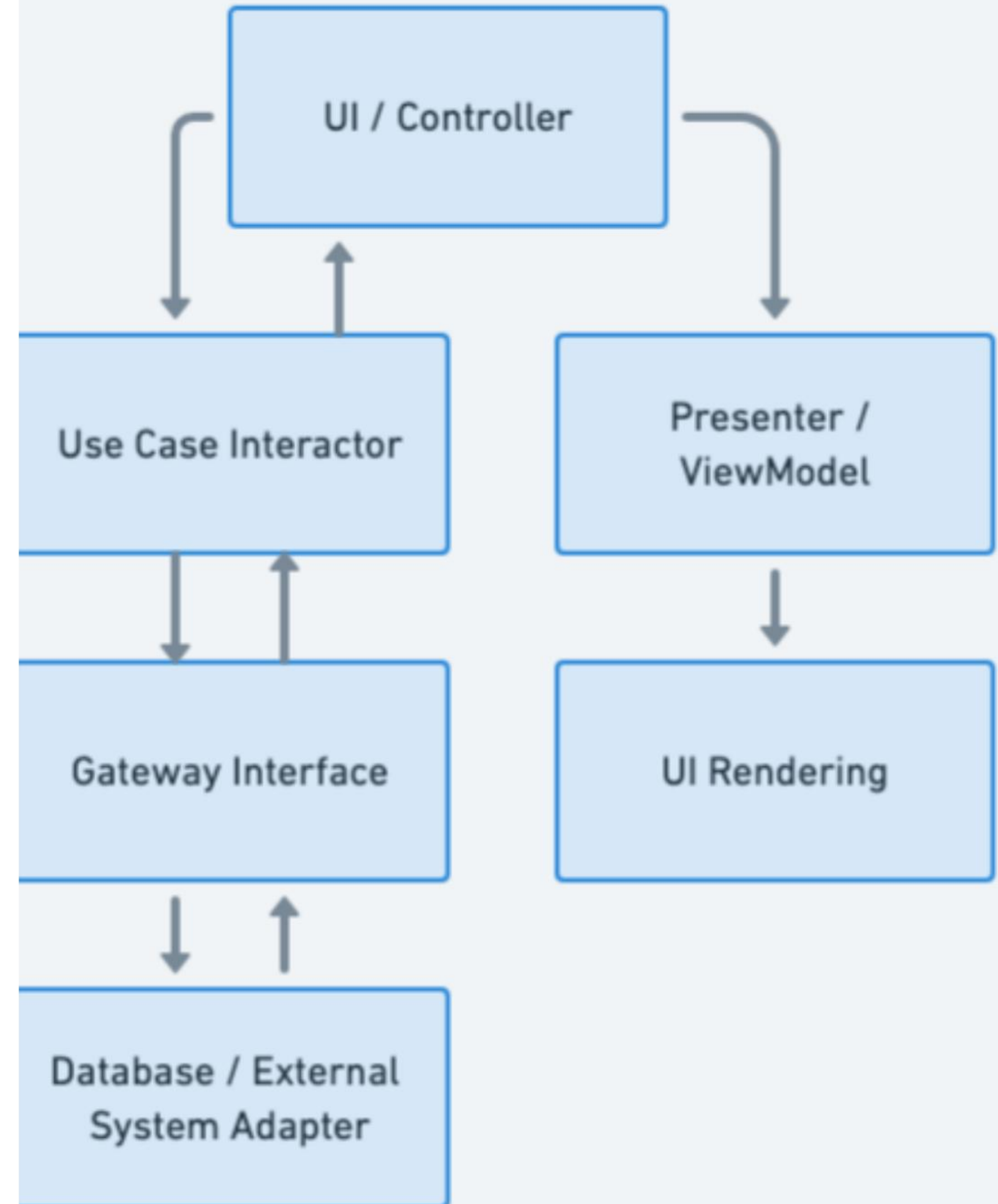
- 📖 **Retrieval-Augmented Generation:** Lewis et al. (2020) established RAG as the industry standard for grounding LLM responses, though primarily focused on synchronous information retrieval.
- 🤖 **LLM-Based Agents:** Xi et al. (2024) surveyed the potential of autonomous agents, highlighting a critical gap in applying structured sales frameworks directly to raw conversational data.
- ⚙️ **Instruction Tuning Limits:** Peng et al. (2023) demonstrated that forcing models to execute complex reasoning simultaneously with dialogue generation drastically increases inference latency.
- 📈 **Sales Automation:** Syam and Sharma (2018) noted that traditional lead scoring relies on rigid heuristics rather than semantic analysis, creating the gap SalesGPT aims to fill.

7. SYSTEM ARCHITECTURE

Decoupled Parallel Agents

The FastAPI orchestration layer securely splits incoming HTTP requests.

- 1. Retrieval & Chat:** Embeds queries via MiniLM, searches Supabase pgvector, and streams Llama-70B responses.
- 2. Background Analysis:** Spawns asynchronous tasks for GPT-OSS-120B (Judge) and Llama-8B (Extractor) to perform heavy compute tasks silently.



8. LIST OF MODULES



Customer Interfaces

React-based frontend Chat Widget with session persistence and the Admin Real-Time Kanban Dashboard.



RAG & Ingestion

Dynamic document chunking, MiniLM embeddings, and pgvector semantic search pipeline.



Agentic Intelligence

The Judge Agent (BANT Scoring), Extractor Agent (Entities), and Email Drafter (Intent analysis).

9. MODULE DESCRIPTION

Judge Agent (BANT Scoring)

Evaluates raw conversation transcripts against four distinct criteria: **Budget, Authority, Need, and Timeline**.

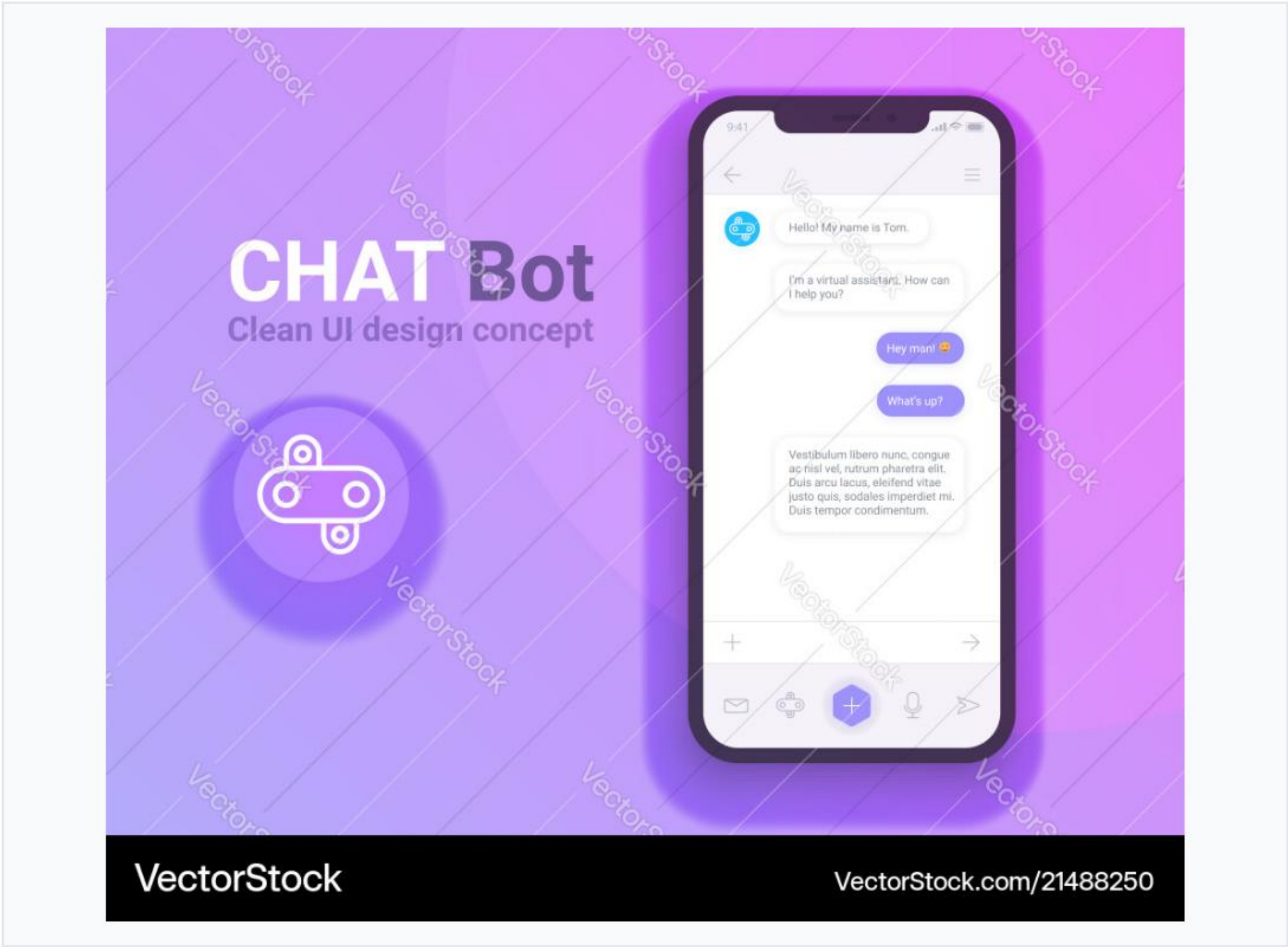
Utilizes a zero-temperature 120B parameter model to ensure deterministic, reproducible scoring from 0 to 100. It dynamically categorizes leads into stages: Visitor, Engaged, Qualified, or Hot Lead.

Extractor & Drafter Agents

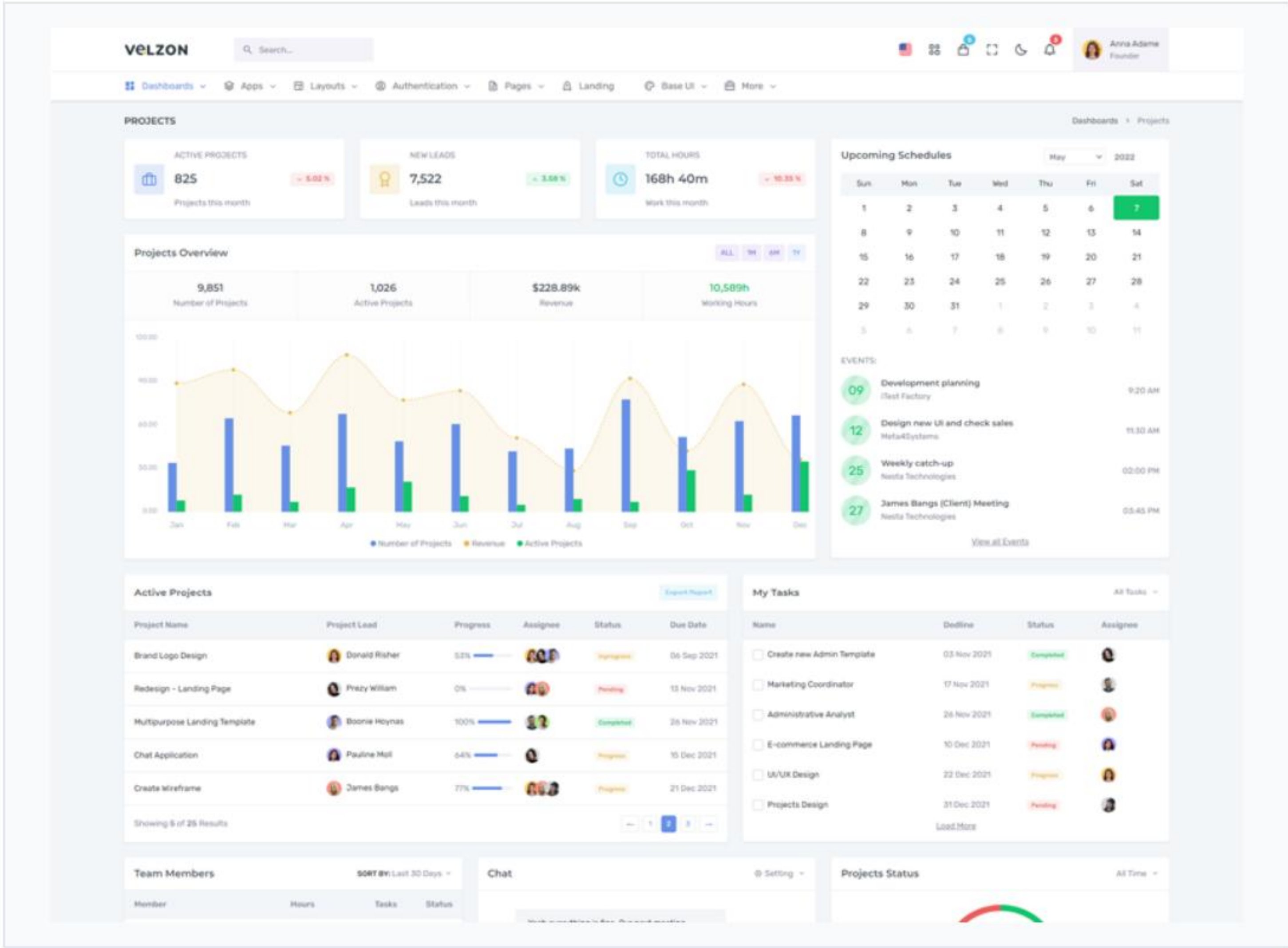
The **Extractor Agent** strictly pulls explicitly mentioned contact data (Name, Email, Role) without hallucination, enriching the Supabase CRM.

The **Email Drafter** identifies the core user intent (e.g., 'pricing_request') and synthesizes a personalized follow-up email grounded entirely in the conversation context.

10. OUTPUT SCREENSHOTS



Fast Track: Customer Chat Widget



Slow Track: Real-Time Admin Kanban Dashboard

IMAGE SOURCES



<https://java-design-patterns.com/assets/img/clean-architecture-flowchart.96d63165.png>

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